C128 USB-C PD Power Adapter v0.1 (DC only, no 50/60 Hz: the CIA TOD clocks will not tick from mains)
Needs a USB-C PD charger offering 15 V (>= 3 A / 45 W recommended). U1 (CH224A) requests 15 V via R1 = 56k.
Q1 connects the 15 V rail only after the PD contract (PG low). U2 and U3 start only above ~13.3 V (EN UVLO).
DIN (square 5-pin): 1 = +5.0 V / -3 A (U2 SY8205 -> U4 eFuse; OVLO 5.64 V, 4.45 A limit; D1 TVS), 3 = +12.3 V DC (U3, D2 TVS) into the C128 "9 VAC" bridge,
2 = GND, 4 = GND/NC, 5 = other 9 VAC leg tied to GND (pin numbers per the Drunk'n'Retro diagram).
VERIFY THE SOCKET PINOUT ON YOUR C128 WITH A METER BEFORE FIRST POWER-UP.

PWR_FLAG PWR_FLAG PWR_FLAG PWR_FLAG PWR_FLAG
VBUS GND VSW 5V_BUCK +12V

VBUS
R2 100k
GATE
GATE C2 100nF/50V VSW
Q1
GATE 1 3 AO3401A
2
VBUS
GATE
R3 100k
PG

J1
VBUS A4
CC1 A5 CC1
CC2 B5 CC2
D- A7
D- B7
D+ A6
D+ B6
SBU1 A8
SBU2 B8
SHIELD SH
GND A1
USB-C (power in)
GND GND

VBUS VBUS VBUS VBUS

VBUS
U1
CC1 7 CC1 VBUS 8 10 PG
CC2 6 CC2 VDD 1
5 DM CFG1 9 CFG1
4 DP CFG2 2
CFG3 3
11
CH224A
GND

VBUS
C1 1uF/50V
GND

CFG1
R1 56k
GND

VSW VSW VSW
C15 10uF/25V C3 10uF/25V C4 100nF/50V
GND GND GND

U2
VSW 7 IN BS 1 BOOT
VSW 8 IN LX 2 SW
EN 3 EN
SS 4 SS FB 5 FB
GND VCC 6 VCC5
9
SY8205FCC
GND

BOOT
C5 100nF/50V
SW
SW
L1 4.7uH
5V_BUCK
5V_BUCK
R6 75k
FB
FB
R7 10k
GND

5V_BUCK 5V_BUCK 5V_BUCK
C6 22uF/25V C7 22uF/25V C18 22uF/25V
GND GND GND

VCC5
C16 1uF/50V
GND

5V_BUCK 5V_BUCK
R13 100k R15 100k
EF_EN EF_OVLO
R14 47k R16 27k
GND GND

5V_BUCK
U4
IN OUT 6 +5V
AUXOFF 3
EN/UVLO FLT 4
OVLO ILM 9 EF_ILM
DVDT GND TIMER 10
8
TPS25947OARPWR
GND

EF_ILM
R17 750R
GND

12V_BUCK
F2 0R
+12V

D1
+5V GND
SMF5.0A

D2
+12V GND
SMF13A

+5V
GND R8 1k D3 LED_A
GREEN
LED_A

J2
GND GND
GND +12V +5V
1
2
3
4
5
C128 POWER (square DIN-5 male)

SS EN
C17 100nF/50V R5 51k
GND GND

VSW
R4 470k
EN

VSW VSW
C9 10uF/25V C10 100nF/50V
GND GND

U3
VSW 3 VIN BOOT 6 BOOT2
EN2 5 EN SW 2 SW2
GND FB 4 FB2
1
TPS54302DDCR
GND

12V_BUCK
C14 100pF
FB2

VSW
R9 470k
FB2
R10 47k
GND

BOOT2
C11 100nF/50V
SW2
SW2
L2 10uH
12V_BUCK
12V_BUCK
R11 100k
FB2
FB2
R12 5.1k
GND

12V_BUCK 12V_BUCK
C12 22uF/25V C13 22uF/25V
GND GND

USB-C PD 15 V -> C128 square DIN-5: +5 V (SY8205) and +12 V DC (TPS54302) on the 9 VAC inp

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File: C128 USB-C PD Power Adapter.kicad_sch
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